Cyber Stalking, Fraud, and Abuse

Joakim Kävrestad

WHO AM I

- Joakim Kävrestad

- ”Lektor” in computer science
  - Teaching focus is forensics and cybersecurity
  - Research focus is user centred security

- Background
  - ”lektor” and ”adjunct” in Skövde since 2016
  - Forensic expert 2014-2016

TODAY

- Some intro to human aspects of cybersecurity
- Cybercrime, cyber aided crime and offline crime
- Fraud and how to avoid it
- Identity theft and how to avoid it
- Cyber abuse
- Online criminal trade

LITTERATURE NOTE!

- Some of the lecture content is not covered by the book but may still appear on the exam
- The law stuff in the book is not written from a Swedish perspective, but I have integrated Swedish perspectives into the slides.

HUMAN ASPECTS

- Truism: User behaviour is a big thing

- Users are expected to adopt a secure behavioour
  - When they don’t, something bad happens
  - We call them the weak link

- Let’s consider the security model that is implicitly or explicitly used by organizations (as it relates to user behaviour)

HUMAN ASPECTS ”SECURITY MODEL”

- Problems
  1. Users passwords are weak
  2. Password re-use
  3. Phishing
  4. Tailgating/piggybacking
  5. Lack of confidentiality
  6. Accidental data leak

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- Solutions
  1. Users should create better passwords
  2. Users should use unique passwords
  3. Users should be cautious about links and attachments
  4. Users should not let other people in
  5. Users should encrypt
  6. User should take care about how they handle print-outs etc

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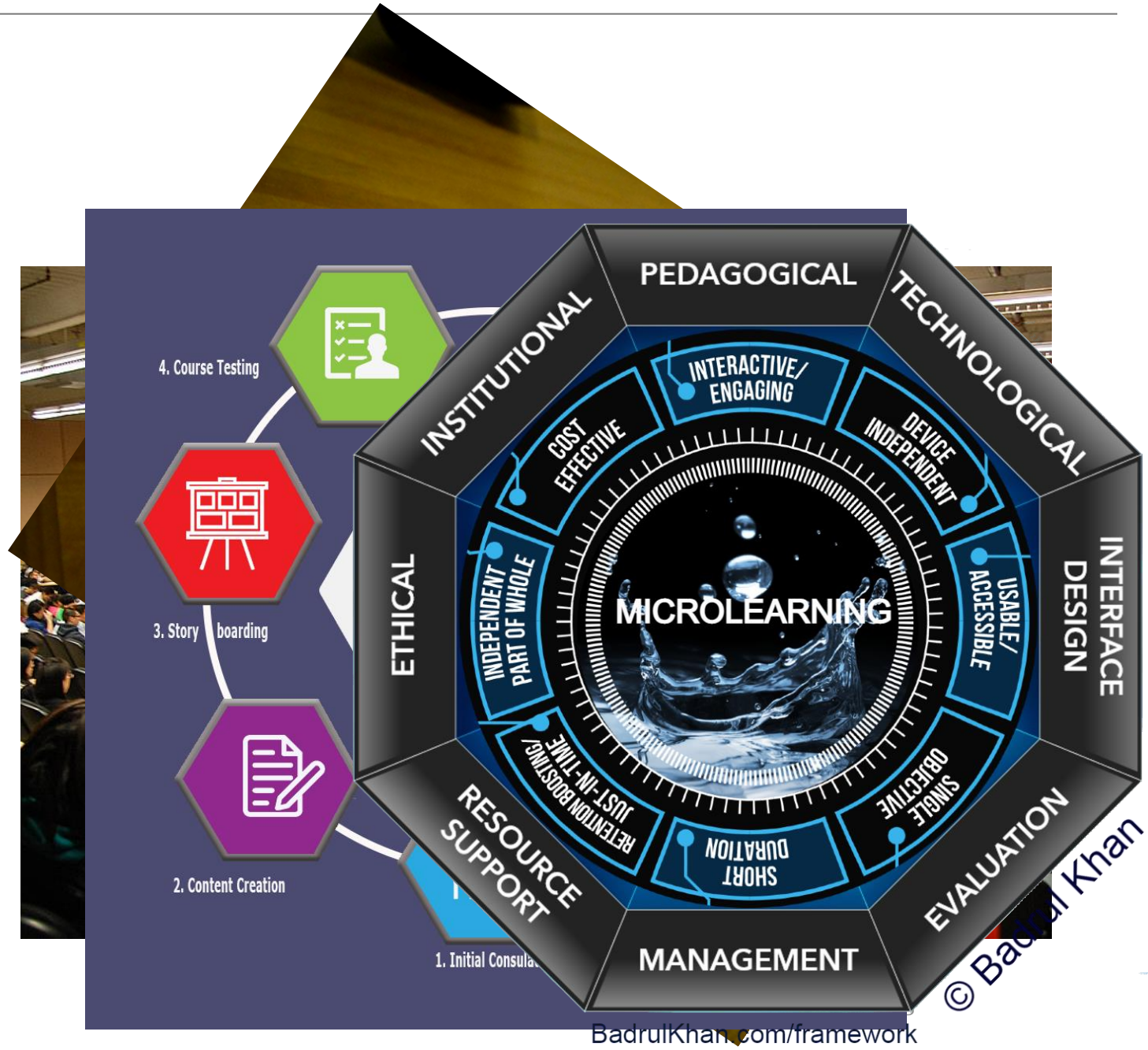
HUMAN ASPECTS ”SECURITY MODEL”

- And **users should** do all of these things correctly and all the time
- If one single user *fails* on one single occasion, we have an incident
- Conclusion: We need users to adopt secure behaviour


4. Course Testing
3. Story boarding
2. Content Creation
1. Initial Consula
PEDAGOGICAL
TECHNOLOGICAL
INTERFACE DESIGN
EVALUATION
MANAGEMENT
RESOURCE SUPPORT
ETHICAL
INSTITUTIONAL
INTERACTIVE/ ENGAGING
DEVICE INDEPENDENT
USABLE/ ACCESSIBLE
SINGLE OBJECTIVE
SHORT DURATION
RETENTION BOOSTING/ JUST-IN-TIME
INDEPENDENT PART OF WHOLE
COST EFFECTIVE
MICROLEARNING
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SOME PROBLEMS WITH TRAINING

- And the problem stays more or less the same

- Some (quite) common problems with training
  - Retention
  - User participation
  - Knowledge != awareness
  - Knowledge != behaviour

EVEN WITH TRAINING…….

*Adopting what is typically defined as secure behaviour is **** hard!*

Reflect:1-2-4-all

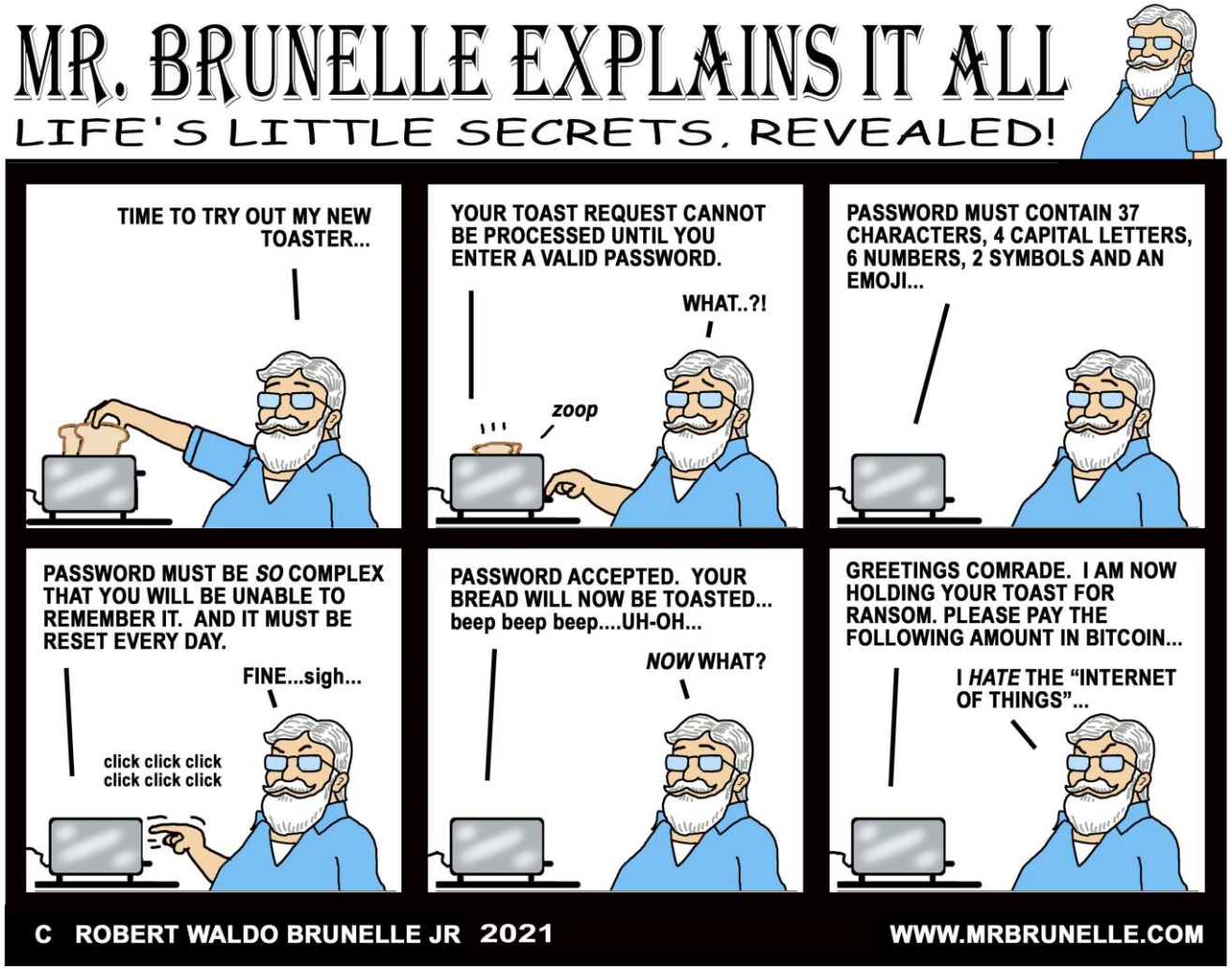
Individually (60 Sec): Write core statements (one-liners) about what you learned in the last lecture part.

Pair up and compare lists (60 Sec) (add new things to your individual lists)

Pair the pairs and repeat (60 Sec)

Then one statement per group of four.

USER BEHAVIOUR

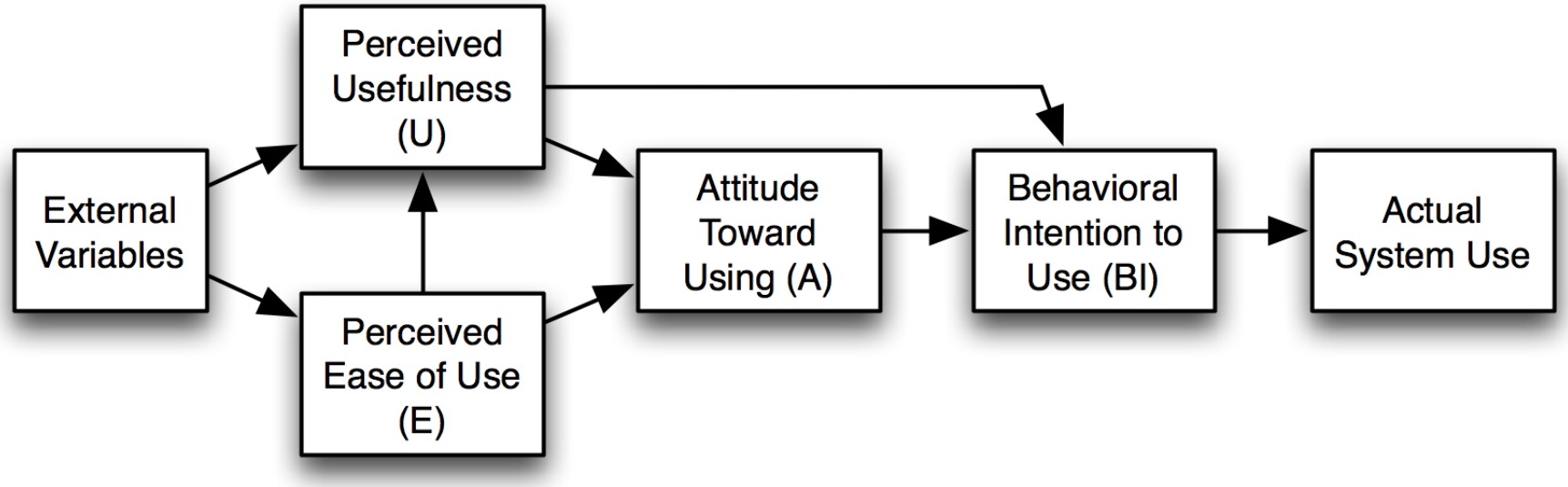


USER BEHAVIOUR

- Consider phishing
  - User are expected not to click evil links or download evil attachements
  - But click all benign links and attachments

- Consider passwords
  - User should create strong passwords
  - Store them safely
  - Any ensure that no one leaks them
  - And what is a strong password anyway?

A MODEL TO EXPLAIN: TAM



WHAT TECHNOLOGY ACCEPTANCE MODEL MEANS

- One way to look at it is that users first decide to adopt (or not adopt) the behaviour we need them to display
- And even then, correct behaviour may not result in a good outcome
- Do we still hold on to the ”user should” security model or try something new?

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CYBERCRIME

- When the world becomes digital, so does crime
- Traditional criminals make use of digital platforms
- New crimes are enabled/created by digital platforms and how they shape society

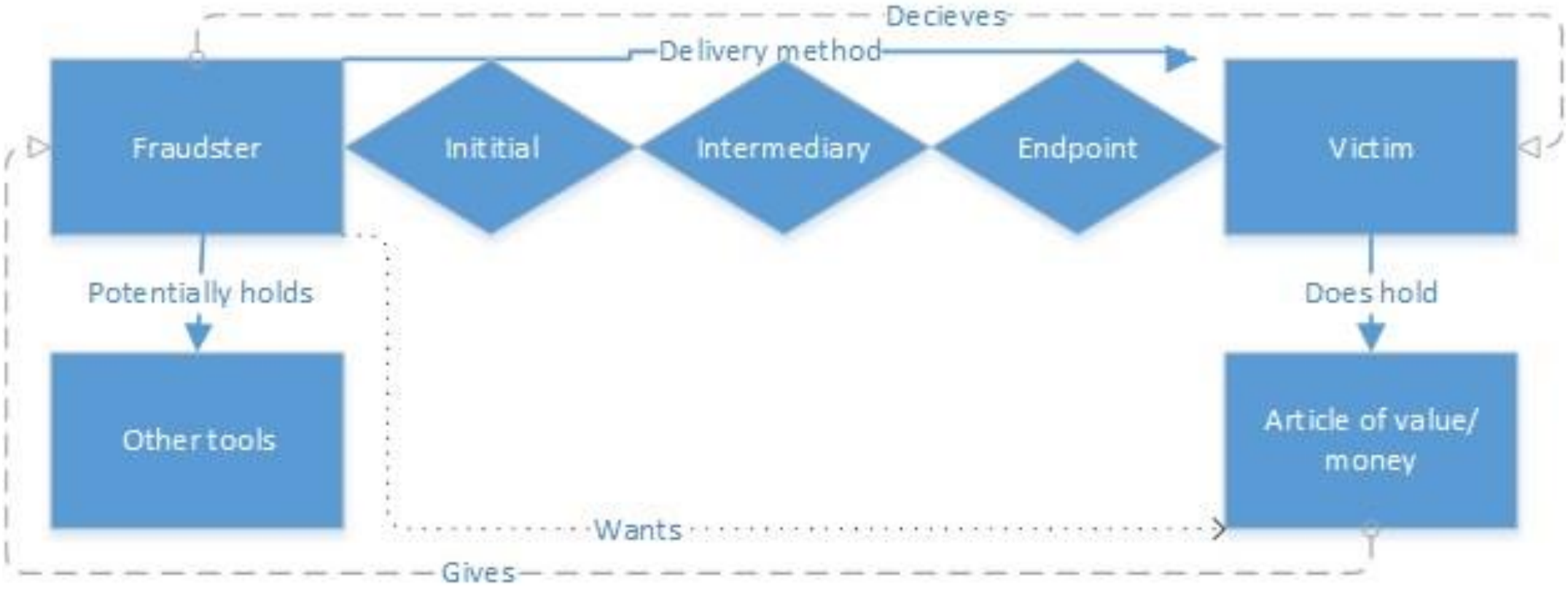
WHAT IS A CRIME

- Defined in law as something illegal
- Typically involves both crimial(s) and victim(s)
- Can be though of in terms of
  - Means
  - Motive
  - Opportunity

CLASSIFYING CYBERCRIME

- Cybercrime
  - Means, Motive, and opportunity is digital
- Cyber-aided crime
  - Some of means motive and opportunity is digital
- Offline crime
  - None of means motive and opportunity is digital

EXAMPLE – (ONLINE) FRAUD



ANOTHER CLASSIFICATION

- ”Private” crimes
  - Fraud, drugs, stalking, sex offenses etc
- Crimes against companies
  - Industrial espionage
- Crimes against states
  - Cyber warfare

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ONLINE FRAUD

- Fraud, in Sweden, is when someone tricks someone else to give up something of monetary value
- Romance scams
- Blocket(ebay) scams
- Nigerian letters
- Fake web store scams

FRAUD – STAY SAFE

- Be cautious!
- Avoid digital payments
- Use trusted payment services
- Dial-back (or similar)

CREDIT CARD SCAMS

- Steal or buy credit card information and use to buy stuff or withdraw money
  - But credit card numbers on darknet or the open web – lets google it!
  - Steal credit cards from individuals
  - The Patrick case

- Skimming

- Stealing from mailboxes

CREDIT CARD SCAMS – STAY SAFE

- Dont share credit card information
- Dont use your credit card with untrusted online stores
- Use safe payment solutions such as Klarna over directly paying by credit card

IDENTITY THEFT

- Use someone identity
- Real identity to take out loans etc
- Social media impersonation to aid other frauds

IDENTITY THEFT – STAY SAFE

- Limit personal information online
- Be careful about how you use digital authentications such as BankID
- Create social media accounts yourself on major social media platforms to ”lock your name”

CYBERSTALKING/BULLYING

- Stalking and bullying in social media
- Sextortion
- Digital abuse/ Non-Consensual Intimate Image Abuse
  - Which is not to be called revenge porn

CYBERSTALKING/BULLYING – STAY SAFE

- Avoid talking to people you dont know IRL
- Know that wats get online stays online
- UK Safe Internet Center
  - StopNCII.org
- To those on the bad side: BEHAVE!

CHILD EXPLOITATION

- Grooming
- Sextortion
- Sexual abuse material

CHILD EXPLOITATION – STAY SAFE

- Maybe more for the parents….
- Let the kids decide over their digital presence
- Talk openly about the hazards and activities online
- Parental control?

ONLINE CRIMINAL TRADE

- You can buy whatever you want online

- Illegal online trade works more or less like legal digital trade
  - Often on Darkweb
  - Using BitCoins

- Lets have a look

ONLINE CRIMINAL TRADE OPEN INTERNET

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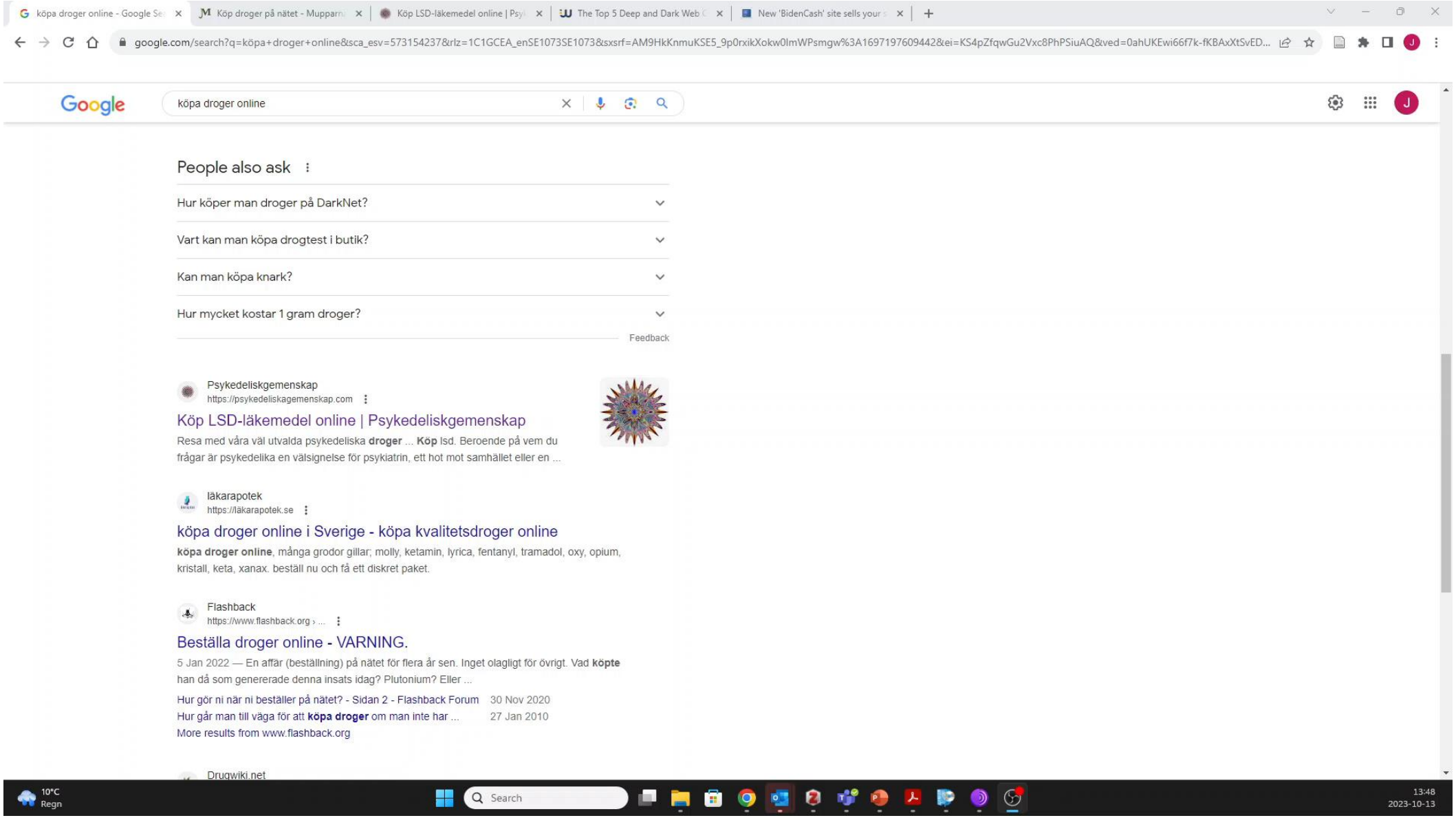

ONLINE CRIMINAL TRADE DARKNET

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Home Market Forum Orders Disputes Messages

Marketplace

Search options

Vendor: All vendors

Category: SIM Cards & Phones

Source: Worldwide

Destination: Worldwide

Min. amount: 10

Unit: All

From: 250 €

To: 750 €

Search term: Indica

Find offers

Categories

Drugs 9016

Popularity Rating Random Quantity Newest

Other > SIM Cards & Phones — Promoted
Swiss Simcards Activated/Anonymous 100chf/pcs
WilhelmTell
Source: Switzerland Ships to: Switzerland
1 - 10 · 104.48 - 10448 €

Other > SIM Cards & Phones
Lyca Sim Karten
Luckyluke123
Source: Germany Ships to: Europe
1 - 50 · 15 - 520 €

Other > SIM Cards & Phones
Anonyme SIM-Karte
CannaComp
Source: Germany Ships to: Germany
1 - 10 · 20 - 150 €

Other > SIM Cards & Phones
Anonymous Sim Card
MisterSwiss

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ONLINE CRIMINAL TRADE – BUYERS END

- Find the store
- Deposit bitcoin
- Browse and buy
  - Leave adress and stuff to the seller
- Get order to your mailbox

ONLINE CRIMINAL TRADE – SELLERS END

- Find the store and create your account
- Collect orders of the day in an excel file
- Pack and post
- Get paid

ONLINE CRIMINAL TRADE – WEAK POINT

- Payment – bitcoins are very traceable

- Physical stuff
  - Mailboxes
  - Customs

- The sellers computers with all buyers information's

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PERSONAL SECURITY HYGIENE

- Good passwords and use multi-factor
- Be a bit suspicious
- Update your devices
- Dont put stuff on the internet if you do not want it there

Questions?

SOURCES

- Course book
- Some optional scientific papers:
- Kävrestad, J., & Nohlberg, M. (2018, September). Defining and Modelling the Online Fraud Process. In *HAISA* (pp. 203-213).
- Kävrestad, J., Furnell, S., & Nohlberg, M. (2021, June). What parts of usable security are most important to users?. In *IFIP World Conference on Information Security Education* (pp. 126-139). Cham: Springer International Publishing.

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